Thyroid Scintigraphy Part ”2” Lec“4”

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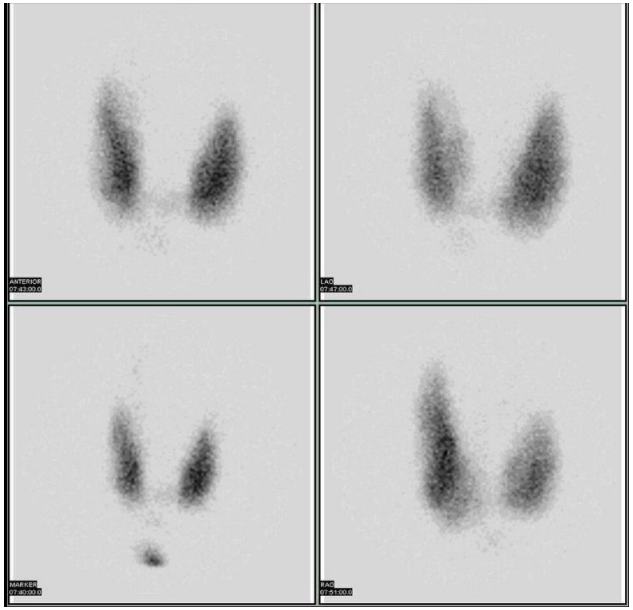
Normal Thyroid Scintigraphy



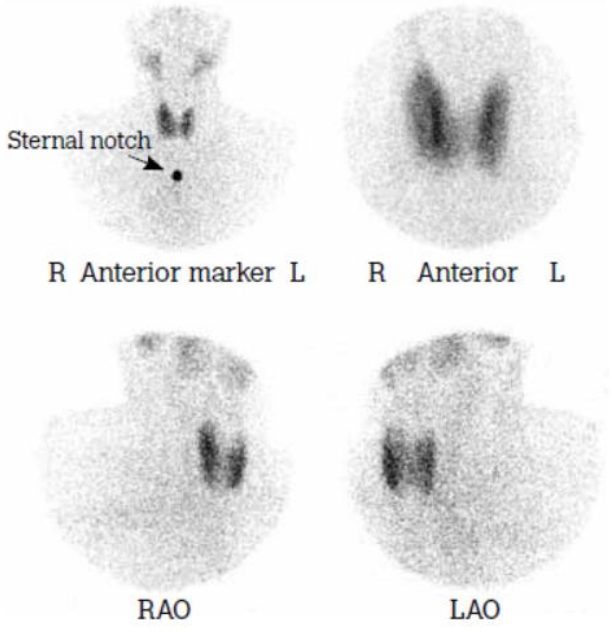
Normal Thyroid Scintigraphy

Typical appearance:

- Butterfly-shaped gland
- Right and left lobes
- Connecting isthmus
- Relatively homogeneous tracer distribution
- Pyramidal lobe may occasionally be visualized
- Normal Anatomical Landmarks



Thyroid scan with Iodine-123

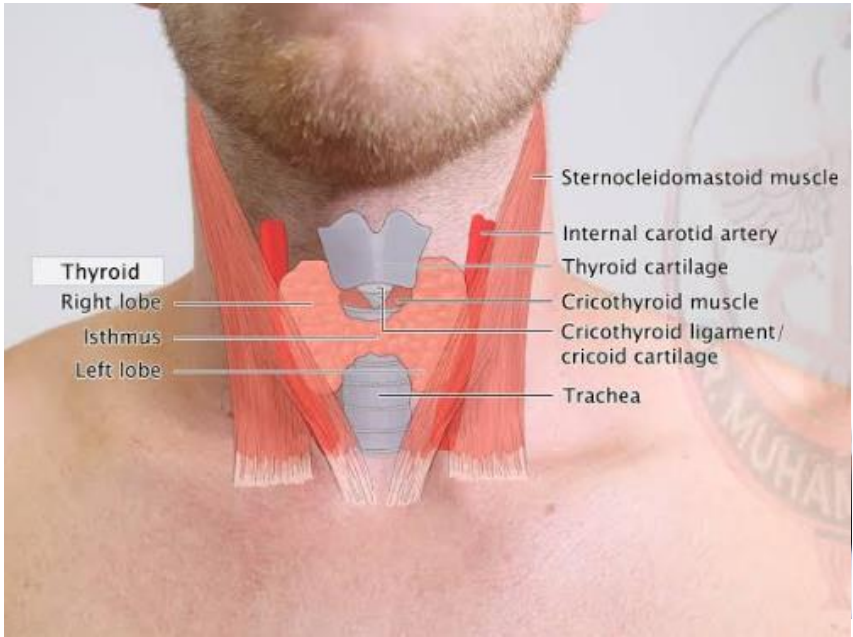


Thyroid scan with TC99M

Normal Anatomical Landmarks

Useful landmarks:

- ✓ Thyroid cartilage
- ✓ Cricoid cartilage
- ✓ Sternal notch
- ✓ Salivary glands
- ✓ A radioactive marker or external anatomical marker can help correlate a palpable abnormality with scintigraphic findings.



Indications for Thyroid Scintigraphy



1. Thyrotoxicosis

- ✓ Graves disease
- ✓ Toxic adenoma
- ✓ Toxic multinodular goiter
- ✓ Thyroiditis
- ✓ Exogenous thyroid hormone
- ✓ Iodine-induced thyrotoxicosis

2. Thyroid nodule with suppressed TSH

- ✓ Hyperfunctioning
- ✓ Isofunctioning
- ✓ Hypofunctioning

3. Detecting Ectopic Thyroid Tissue:

To locate functional thyroid tissue outside its normal position (such as a lingual or substernal/retrosternal thyroid)

4. Diagnosing Thyroiditis:

To help differentiate between various forms of thyroid inflammation (such as subacute thyroiditis, which typically shows very low uptake).

5. Evaluating Pediatric and Congenital Conditions:

To assess congenital hypothyroidism or juvenile thyroid disorders.

6. Post-Treatment or Surgical Monitoring:

To check the status of the gland or remnant tissue after thyroid surgery or radioiodine therapy.



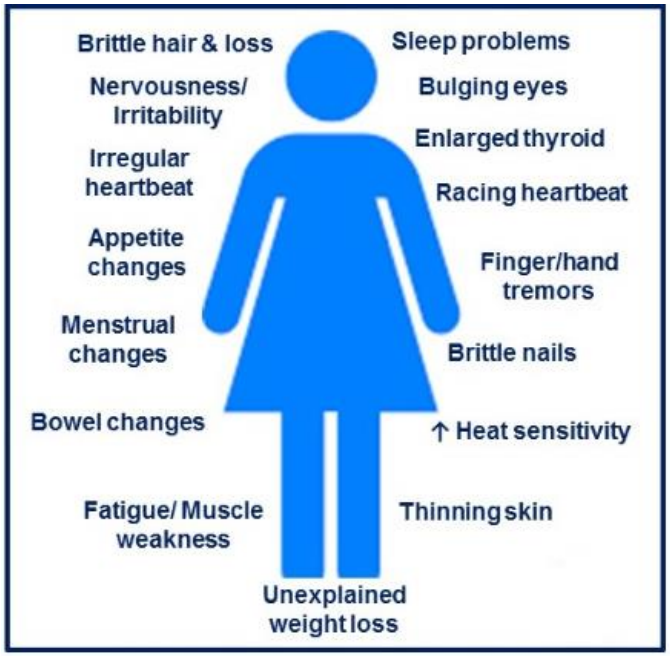
Major indications

1. Thyrotoxicosis

Particularly when the cause needs to be determined.

Examples:

- ✓ Graves disease
- ✓ Toxic adenoma
- ✓ Toxic multinodular goiter
- ✓ Thyroiditis
- ✓ Exogenous thyroid hormone
- ✓ Iodine-induced thyrotoxicosis



• **Hyperthyroidism:** The thyroid gland makes and releases too much hormone (an overactive thyroid).

• **Thyrotoxicosis:** The general clinical effect of increased thyroid hormone in patient tissues.



Graves Disease

Typical scintigraphic pattern:

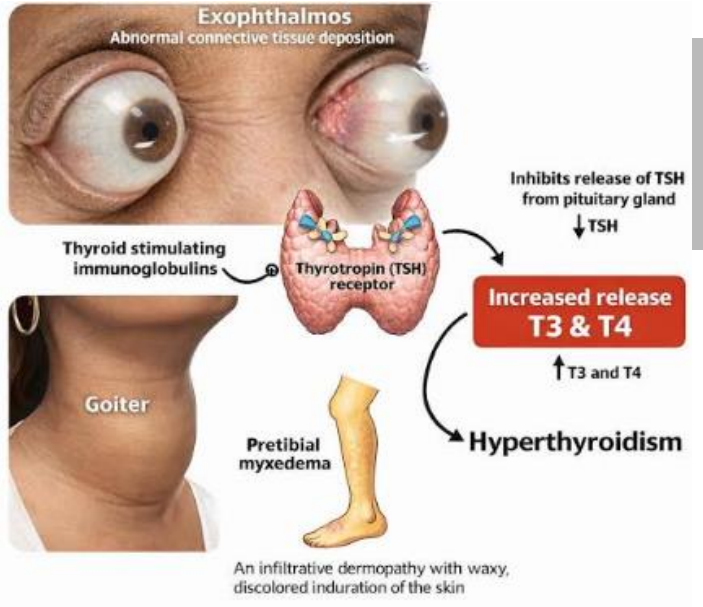
- ❑ Diffuse increased uptake
- ❑ Enlarged gland
- ❑ Homogeneous distribution
- ❑ Increased activity

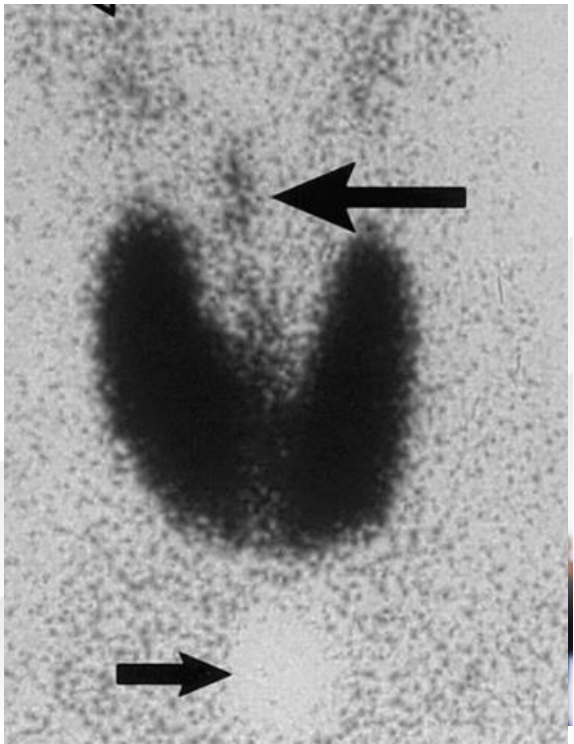
Often visualization of both lobes and isthmus

Typical pattern: "Diffuse toxic goiter"

Mechanism

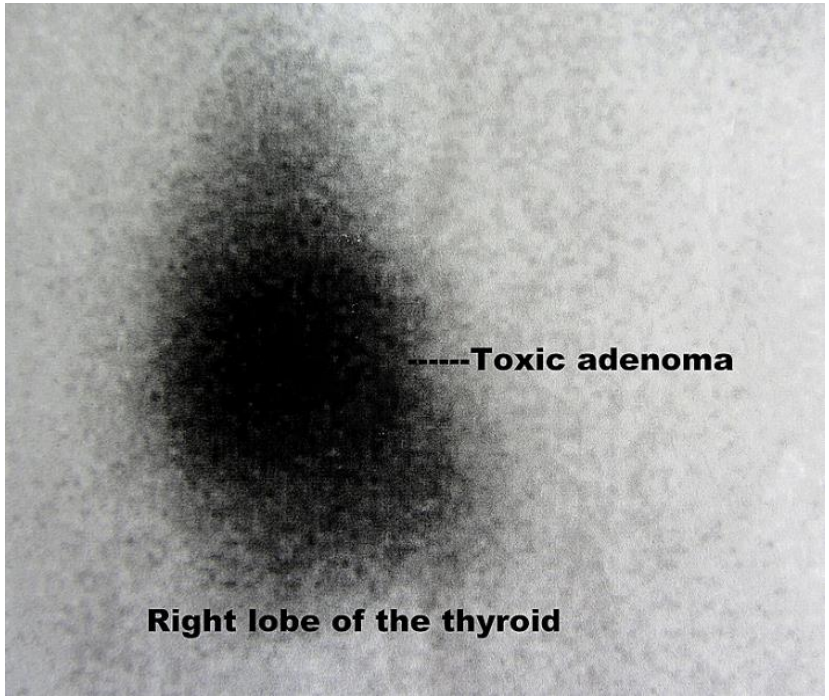
- ❑ TSH-receptor stimulation
- ❑ increased NIS activity
- ❑ increased iodine trapping
- ❑ increased hormone synthesis
- ❑ increased radionuclide uptake.

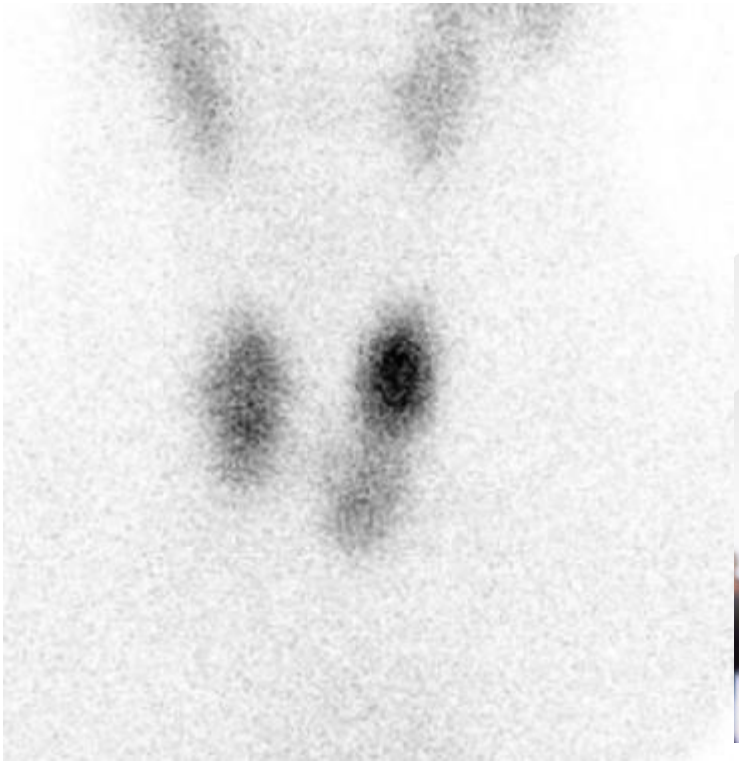




Toxic Adenoma

Typical appearance:
Single focal hot nodule
with suppression of the remaining thyroid tissue.
Pattern:
The autonomous nodule functions independently of normal TSH regulation.





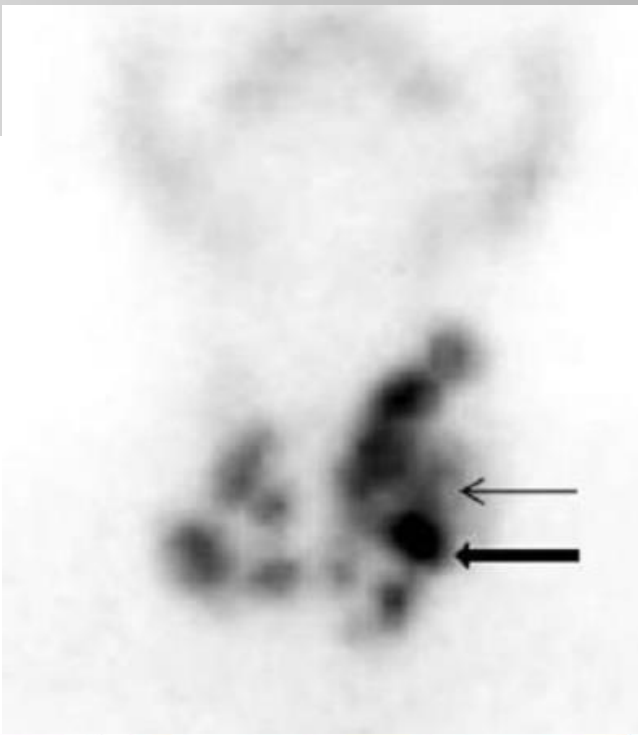
Toxic Multinodular Goiter

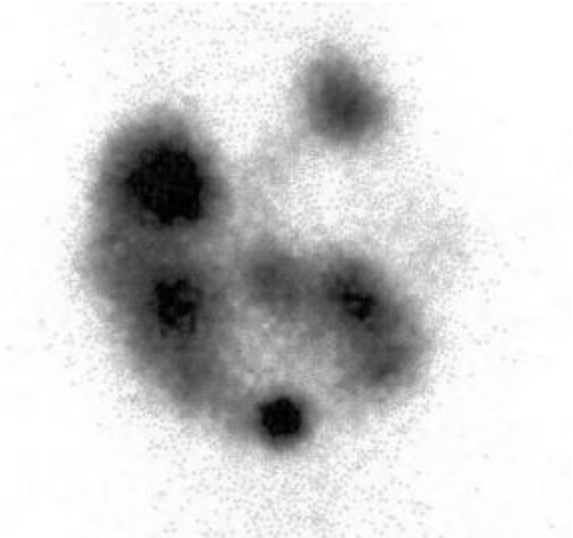
Typical pattern:

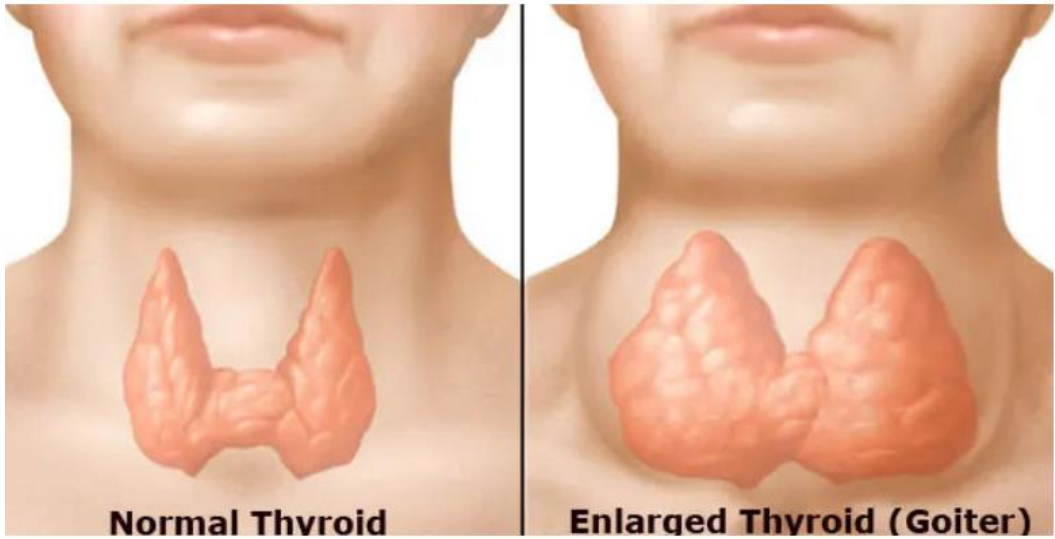
- ✓ Multiple areas of increased uptake
- ✓ Multiple areas of reduced uptake
- ✓ Heterogeneous distribution
- ✓ Suppression of some normal thyroid tissue

Appearance:

"Patchy / heterogeneous uptake"







Thyroiditis

Examples:

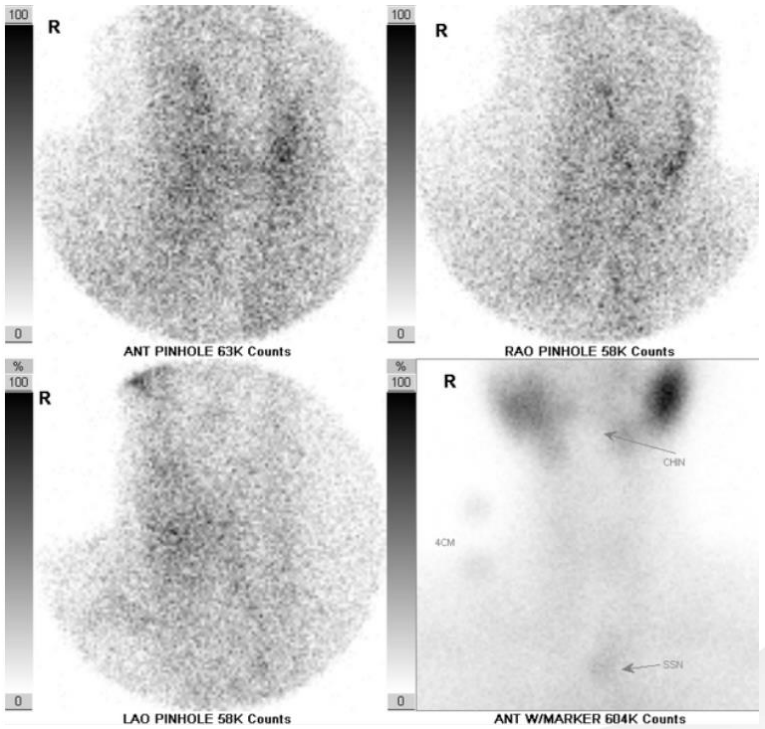
- ❑ Subacute thyroiditis
- ❑ Painless thyroiditis
- ❑ Postpartum thyroiditis

Typical pattern during thyrotoxic phase:

- ❑ Very low thyroid uptake

because hormone excess is due predominantly to release of preformed hormone rather than increased synthesis.

This is an important distinction from Graves disease.





Simplified comparison

Condition	Uptake
Graves	↑↑ diffuse
Toxic adenoma	↑ focal
Toxic MNG	↑ heterogeneous
Thyroiditis	↓
Exogenous thyroid hormone	↓

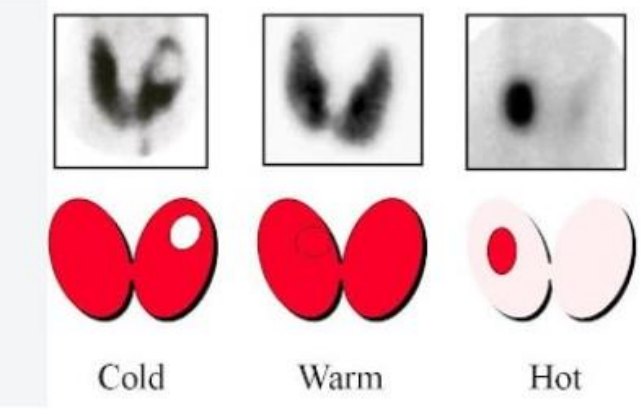


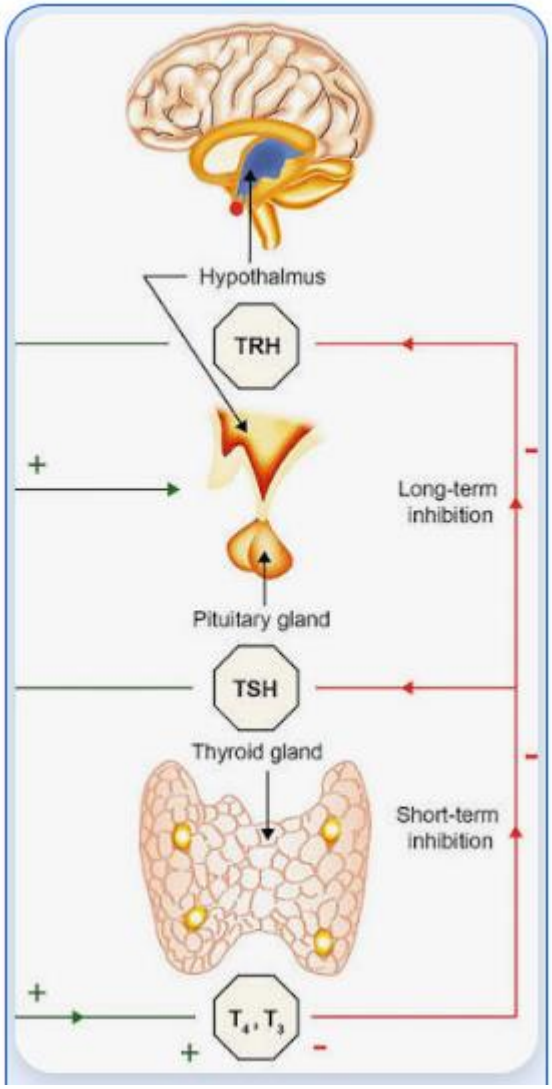
2. Thyroid nodule with suppressed TSH

This is an important indication.
If: TSH ↓
→ autonomous thyroid tissue should be considered.

Scintigraphy can determine whether the nodule is:

1) Hyperfunctioning
2) Isofunctioning
3) Hypofunctioning





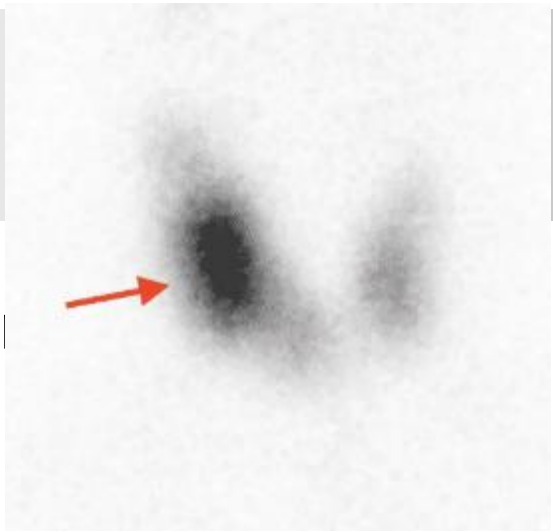
Thyroid Nodules:

Hot nodule

Increased tracer uptake.

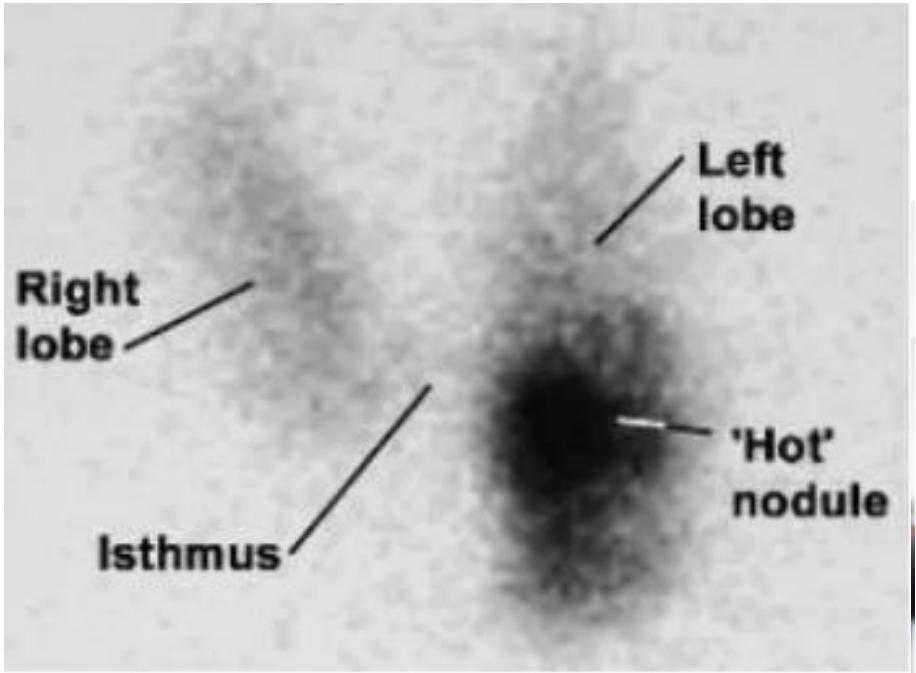
Usually represents:

- ✓ autonomously functioning thy tissue
- ✓ A hyperfunctioning nodule is generally less likely to be malignant than a nonfunctioning nodule, although scintigraphy does not by itself exclude malignancy.









Cold nodule

Reduced or absent tracer uptake.

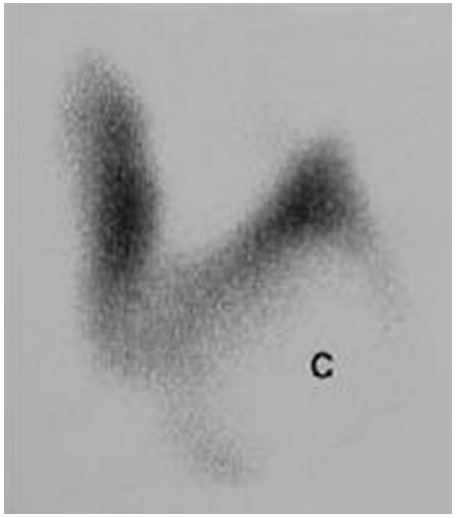
Important:

Cold ≠ cancer.

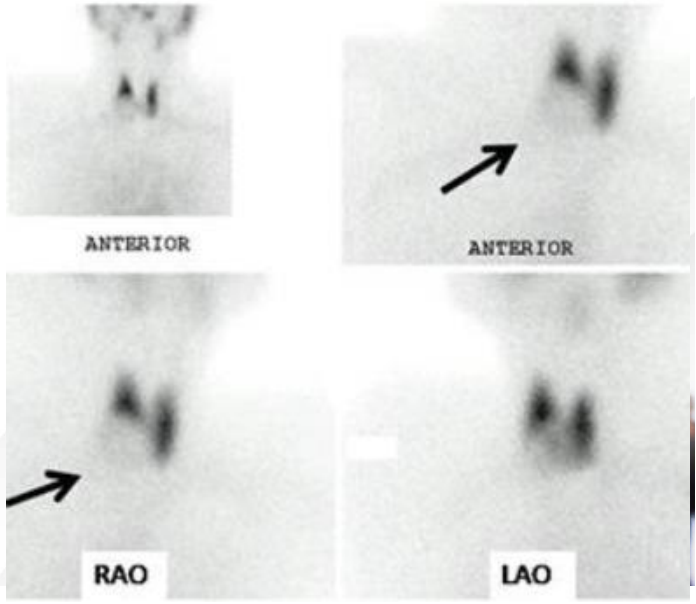
Most cold nodules are benign, but they may require evaluation according to ultrasound risk stratification and clinical context.

Ultrasound remains the principal anatomical examination for structural thyroid-nodule assessment.







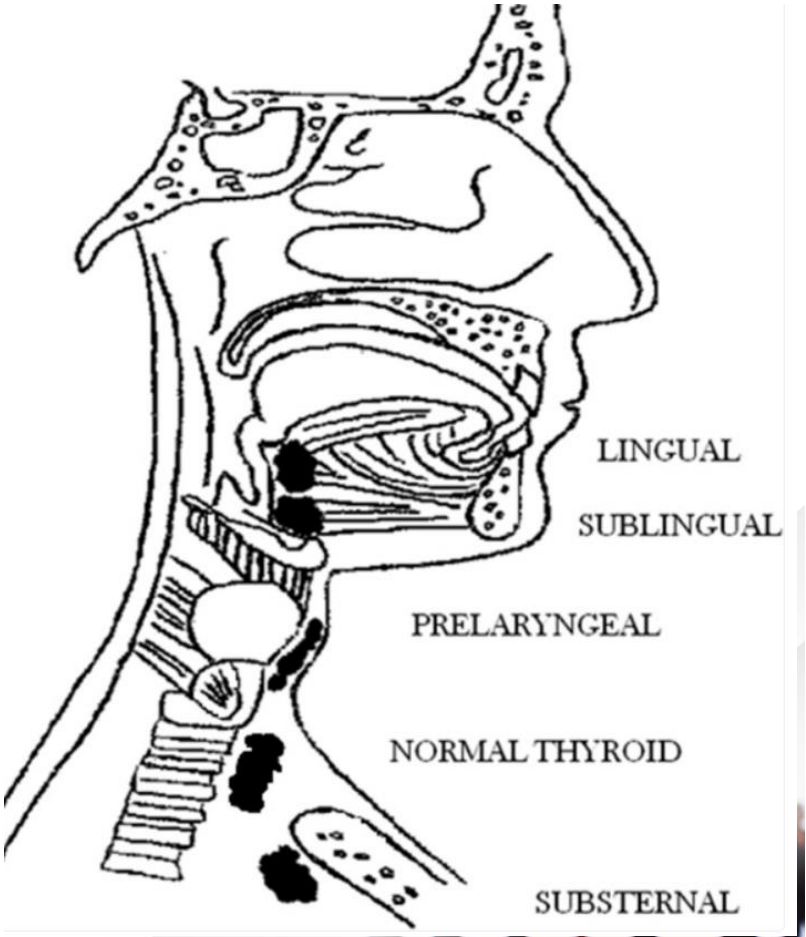


3. Ectopic Thyroid Tissue



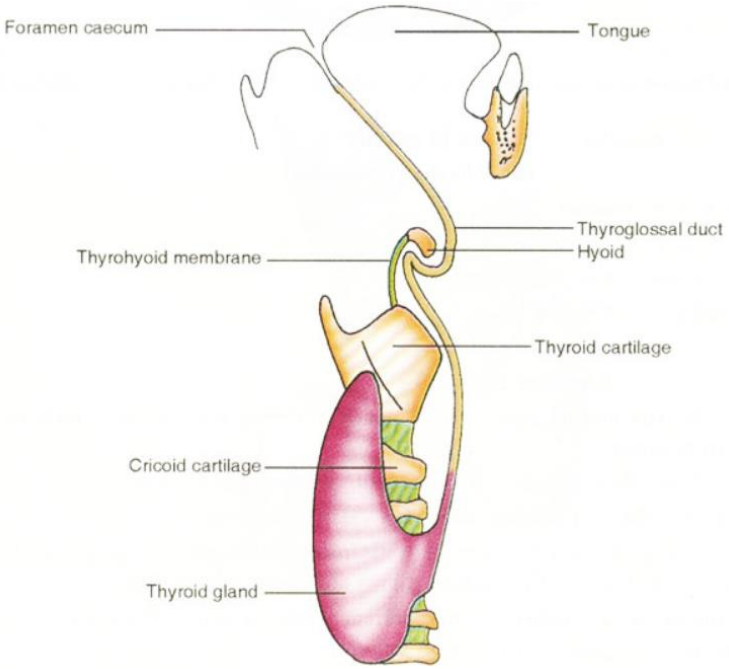
An ectopic thyroid gland is one which is located in a location other than the normal position anterior to the laryngeal cartilages.

1. lingual: base of tongue (90%)
2. Sublingual: below the tongue
3. Prelaryngeal
4. Other sites, e.g. mediastinum (<1%), intratrachea



Variant anatomy

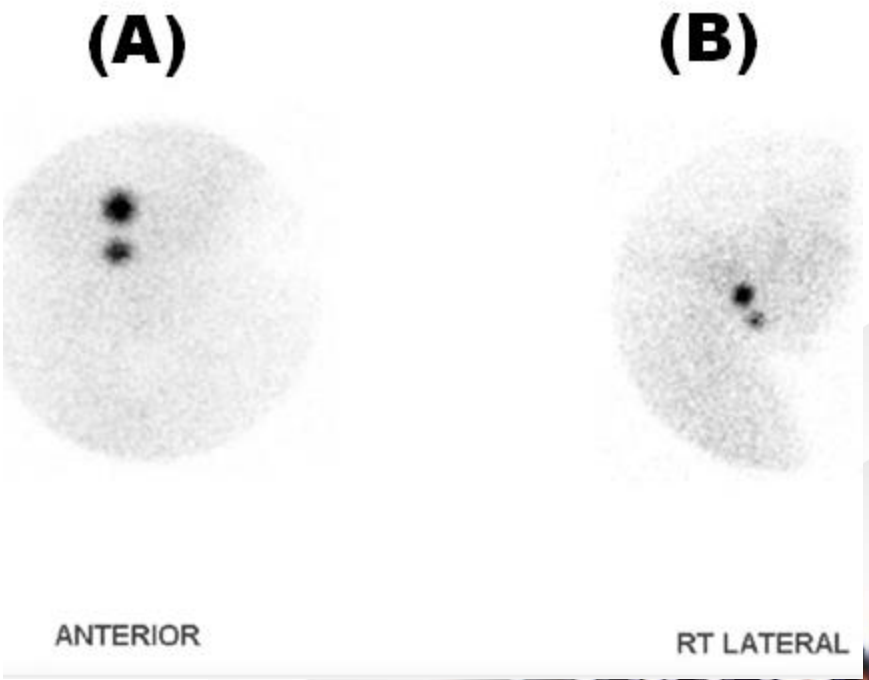
During embryological development, the thyroid gland migrates down from the foramen cecum at the posterior aspect of the tongue to its permanent location in the infrahyoid neck. This normal migration can be halted at any point, or indeed can go 'off-target' with thyroid tissue coming to rest in unusual locations within the neck or elsewhere.



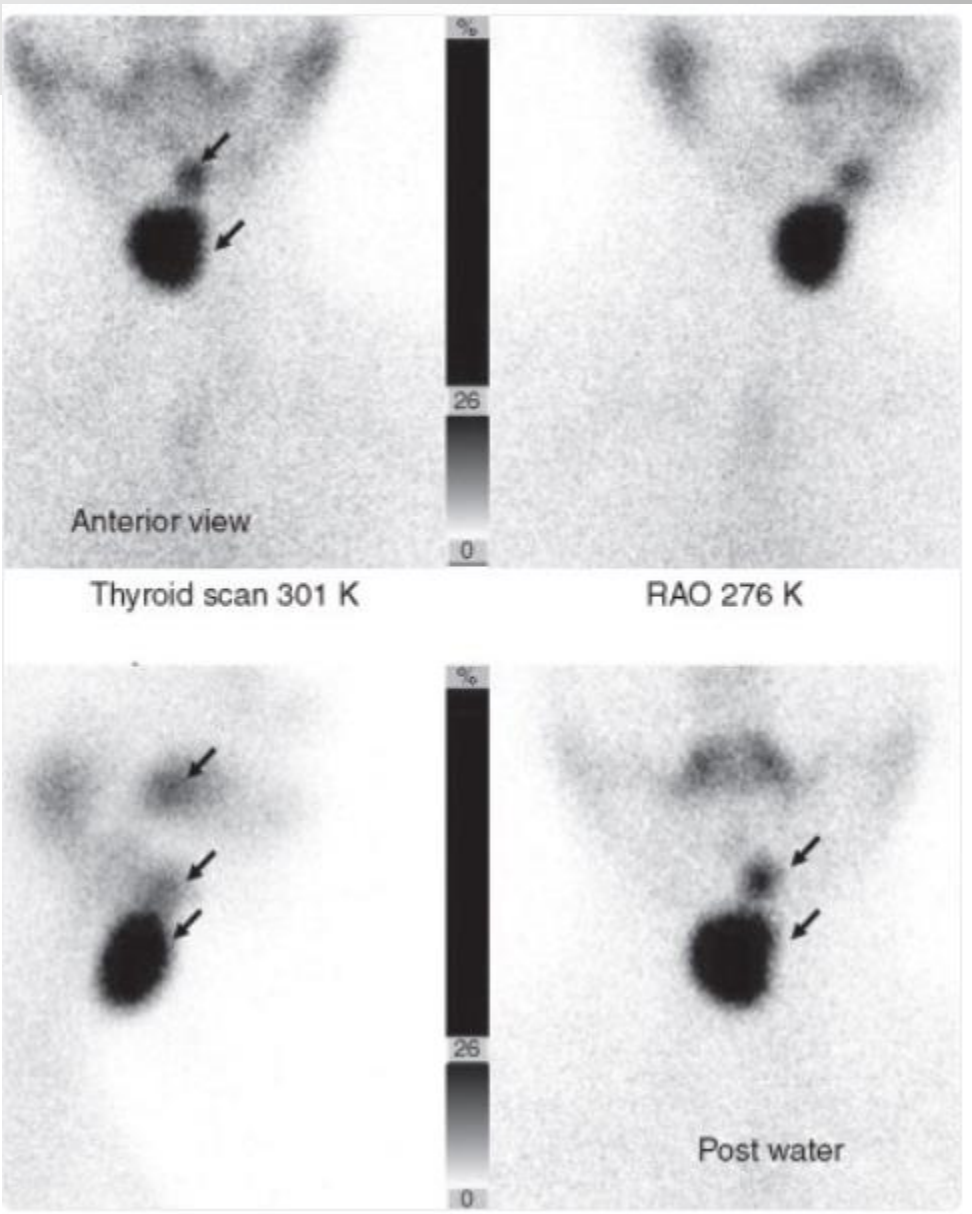


Ectopic thyroid is a rare condition resulting from aberrant embryological development, primarily involving abnormal migration of thyroid tissue. The lingual region is the most common site, and diagnostic methods include US, CT, and MRI, with thyroid scintigraphy considered the gold standard.

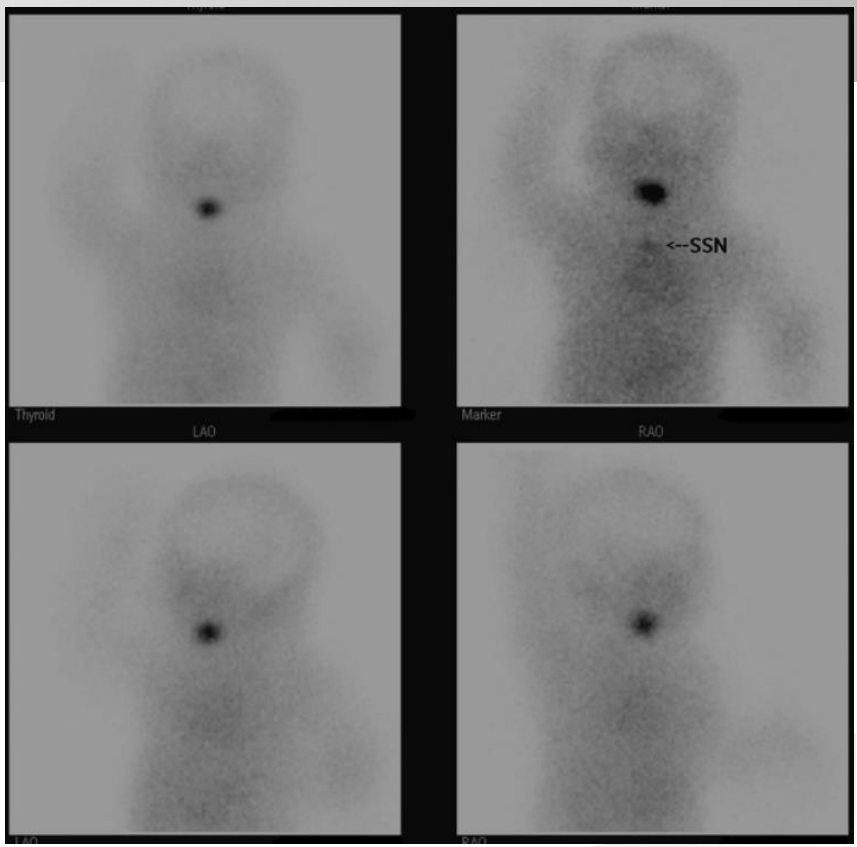
(A) Anterior and (B) lateral projection pinhole images from a technetium-99m pertechnetate thyroid scan showing 2 foci of uptake in the upper and lower neck at midline. There was no evidence of a normal thyroid gland.



Technetium 99m thyroid scan revealed tracer uptake at three different levels other than the normal thyroid gland region.



Tc-99m pertechnetate was administered intravenously and planar images in anterior, left anterior oblique, and right anterior oblique projections were acquired 20 minutes post injection. Well-circumscribed area of radiotracer concentration in the posterior floor of the mouth representing lingual thyroid. No radiotracer uptake is seen in the anatomical region of the thyroid gland. Thyroid uptake function=1% (normal: 0.4–4%).



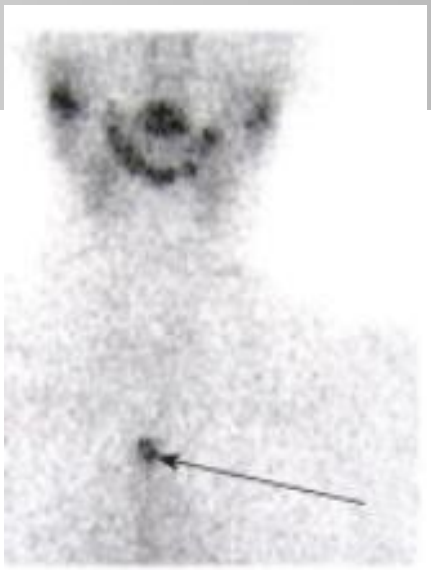


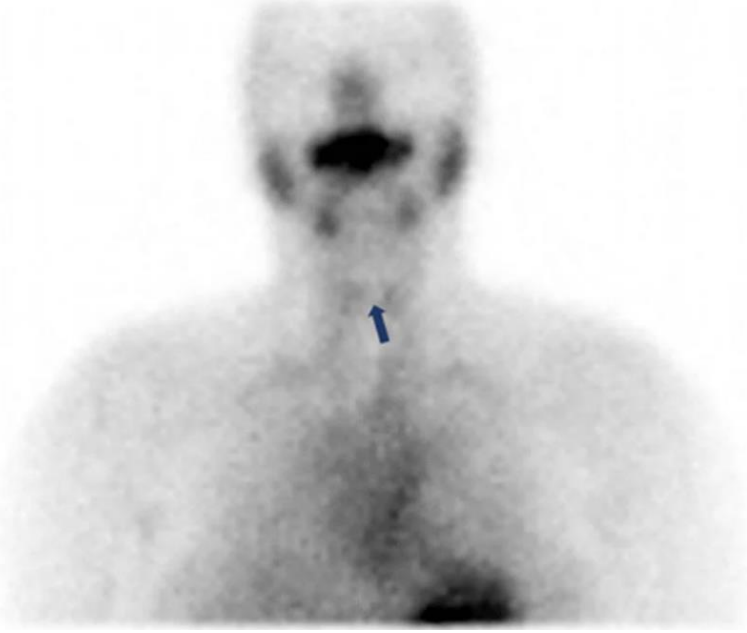
4. Thyroiditis



Thyroiditis

Thyroiditis, demonstrating absent uptake in expected position of thyroid gland, and relatively prominent uptake in salivary glands.





Thyroid scan showing thyroiditis. The arrow shows decreased uptake of the radio indicator (Technetium) within the thyroid gland, suggestive of thyroiditis

5. Evaluating Pediatric and Congenital Conditions



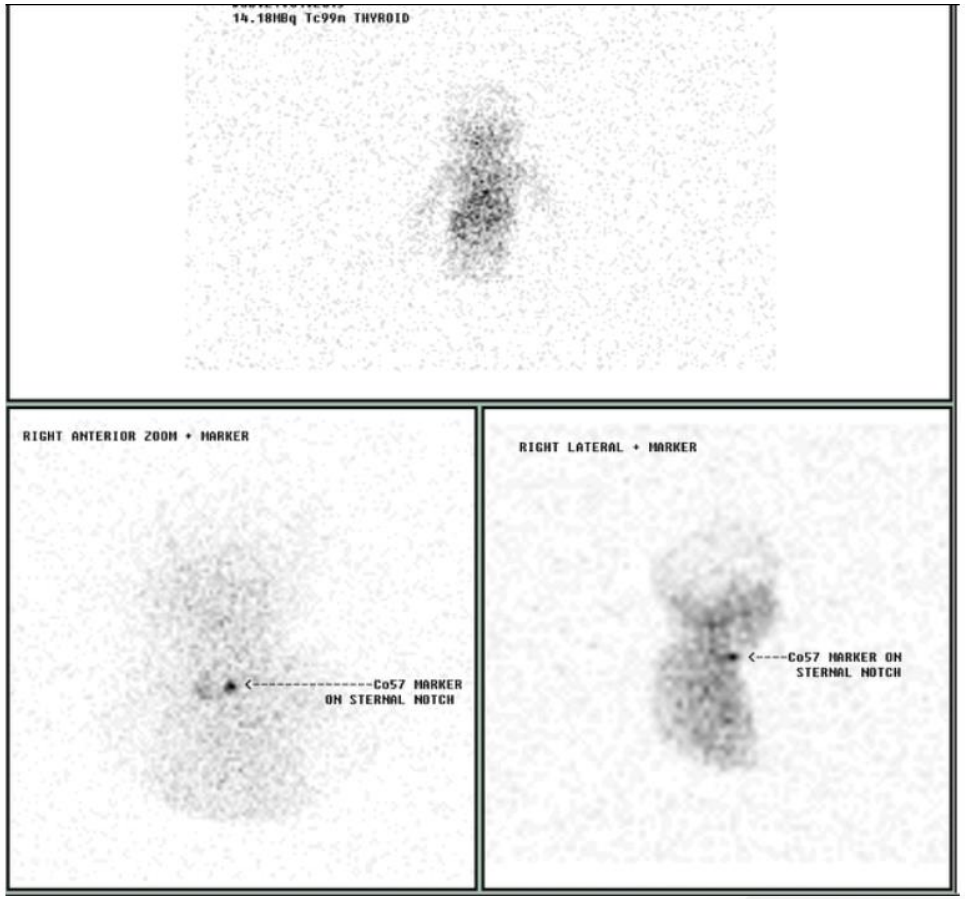
- ✓ In case of thyroid hormone deficiency which presented at birth. Thyroid hormones are essential for brain development and normal cognitive function.

- ✓ Common symptoms of congenital hypothyroidism (CH) include constipation, decreased activity, increased sleep and feeding difficulty. Common signs include dry skin, macroglossia and umbilical hernia.

- ✓ If congenital hypothyroidism is left untreated after birth, it can lead to permanent intellectual disability and growth failure.



Tc-99m pertechnetate was administered intravenously.
Sternal notch marker noted.
There was no isotope uptake demonstrated in the cervical region or tongue base. In this clinical appearances are most likely to represent thyroid gland agenesis.





6. Post-Treatment or Surgical Monitoring

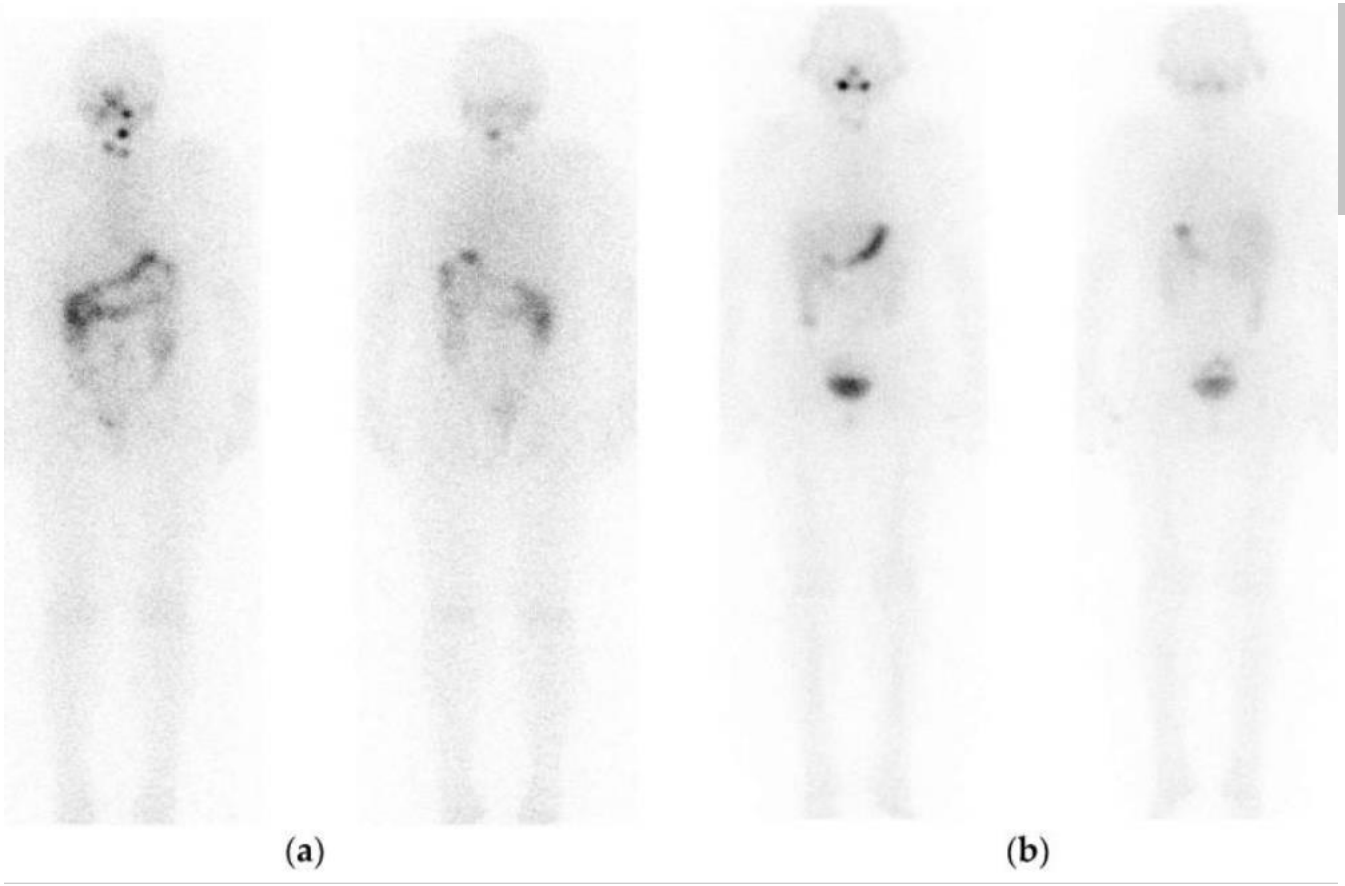


I-131 has been incorporated into the management of differentiated thyroid cancer (DTC) for many years. Following total thyroidectomy, the use of I-131 has been traditionally used as an important part of the initial DTC treatment algorithm. Over time, the role of postoperative I-131 ablation has changed.

The postoperative radioiodine imaging of DTC includes pre-treatment diagnostic I-131, whole-body imaging, and post-I-131 treatment whole-body imaging.







A differentiated thyroid cancer (DTC) patient after total thyroidectomy. (**a**) planar I-131, whole-body images, anterior and posterior plane , prior to I-131 therapy (**b**) Post treatment WBI, planar images, anterior and posterior plane. High radioiodine uptake is seen in 3 foci in the thyroid bed, which is significantly reduced on the post treatment I-131, whole-body images.



Radiation Protection

1) ALARA

As Low As Reasonably Achievable

2) Principles:

❑ Time

Minimize unnecessary exposure time.

❑ Distance

Maximize distance from radioactive sources.

❑ Shielding

Use appropriate shielding and barriers.

3) Contamination control

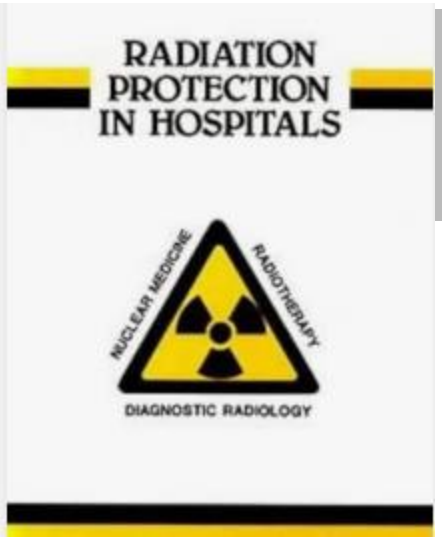
Prevent contamination of:

Work surfaces Camera

Patient Staff

Clothing

4) Technologist Safety





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Thank You



